

A Modes of Convergence

Let $\{Z_n\}_{n=1}^{\infty}$ be a sequence of scalar random variables and let Z be a scalar random variable. A constant is also allowed as the limit.

Definition 1 (Convergence in Probability). We say that Z_n *converges in probability* to Z , written $Z_n \xrightarrow{p} Z$, if for every $\varepsilon > 0$,

$$\Pr(|Z_n - Z| > \varepsilon) \rightarrow 0 \quad \text{as } n \rightarrow \infty. \quad (1)$$

Definition 2 (Convergence in Mean Square). We say that Z_n *converges in mean square* to Z , written $Z_n \xrightarrow{ms} Z$, if

$$\mathbb{E}[(Z_n - Z)^2] \rightarrow 0, \quad (2)$$

provided that the variables have finite second moments.

Proposition 1. If $Z_n \xrightarrow{ms} Z$, then $Z_n \xrightarrow{p} Z$.

Proof. For any $\varepsilon > 0$, Markov's inequality applied to the nonnegative random variable $(Z_n - Z)^2$ gives

$$\Pr(|Z_n - Z| > \varepsilon) = \Pr((Z_n - Z)^2 > \varepsilon^2) \leq \frac{\mathbb{E}[(Z_n - Z)^2]}{\varepsilon^2} \rightarrow 0.$$

This is the definition of convergence in probability. ■

Remark 1 (Bias-Variance Decomposition). For a constant target c , the same result follows from the bias-variance decomposition:

$$\mathbb{E}[(Z_n - c)^2] = \mathbf{Var}(Z_n) + (\mathbb{E}(Z_n) - c)^2. \quad (3)$$

If variance and squared bias both vanish, the **mean squared error (MSE)** tends to zero. Mean square convergence then implies convergence in probability by **Proposition 1**.

Example 1 (Converse does not hold). Let $\{Z_n\}$ be a sequence of random variables with $\text{supp}(Z_n) = \{0, n\} \forall n$. Let $Z_n = n$ with probability $1/n$ and $Z_n = 0$ otherwise. For every fixed $\varepsilon > 0$ and sufficiently large n ,

$$\Pr(|Z_n| > \varepsilon) = n^{-1} \rightarrow 0$$

Thus $Z_n \xrightarrow{p} 0$. However, since $\mathbb{E}(Z_n^2) = n$, the sequence does not converge to zero in mean square.

Definition 3 (Convergence in Distribution). Let F_{Z_n} and F_Z be the **cumulative distribution function (CDF)** of Z_n and Z respectively. We say that Z_n *converges*

in distribution to Z , written $Z_n \rightsquigarrow Z$, if

$$F_{Z_n}(z) \rightarrow F_Z(z) \quad \text{for all } z \in \mathbb{R} \text{ at which } F_Z(z) \text{ is continuous.} \quad (4)$$

Remark 2. One technical difference between convergence in probability and convergence in distribution is that the latter does not require the random variable to be defined on the same probability space, as we only require the **CDFs** to converge at continuous points.

Example 2. Consider the sequence of random variables Z_n with **CDFs**

$$F_{Z_n}(z) = \begin{cases} 0, & z < -\frac{1}{n} \\ \frac{1}{2} + \frac{n}{2}z, & z \in \left[-\frac{1}{n}, \frac{1}{n}\right] \\ 1, & z > \frac{1}{n} \end{cases} \quad (5)$$

It is easy to see that $F_{Z_n}(z) \rightarrow F_Z(z)$ for all $z \neq 0$, where F_Z is the **CDF** of the constant random variable $Z = 0$. Note that $F_{Z_n}(0) = 1/2 \neq F_Z(0) = 1$.

Proposition 2. If $Z_n \xrightarrow{p} Z$, then $Z_n \rightsquigarrow Z$.

Proof. Fix a point z at which F_Z is continuous and let $\delta > 0$. If $|Z_n - Z| \leq \delta$, then $Z \leq z - \delta$ implies $Z_n \leq z$, and $Z_n \leq z$ implies $Z \leq z + \delta$. Consequently,

$$F_Z(z - \delta) - \Pr(|Z_n - Z| > \delta) \leq F_{Z_n}(z) \leq F_Z(z + \delta) + \Pr(|Z_n - Z| > \delta).$$

Since $Z_n \xrightarrow{p} Z$, the probability terms tend to zero, giving

$$F_Z(z - \delta) \leq \liminf_{n \rightarrow \infty} F_{Z_n}(z) \leq \limsup_{n \rightarrow \infty} F_{Z_n}(z) \leq F_Z(z + \delta).$$

Letting $\delta \rightarrow 0$, continuity at z makes both outer bounds converge to $F_Z(z)$. Thus $F_{Z_n}(z) \rightarrow F_Z(z)$ at every continuity point of F_Z . ■

Proposition 3. If $Z_n \rightsquigarrow c$ for a constant c , then $Z_n \xrightarrow{p} c$.

Proof. The **CDF** of the constant c is zero below c and one at or above c . For any $\varepsilon > 0$, both $c - \varepsilon$ and $c + \varepsilon$ are continuity points of this **CDF**. Convergence in distribution therefore gives $F_{Z_n}(c - \varepsilon) \rightarrow 0$ and $F_{Z_n}(c + \varepsilon) \rightarrow 1$. Hence,

$$\begin{aligned} \Pr(|Z_n - c| > \varepsilon) &= \Pr(Z_n < c - \varepsilon) + \Pr(Z_n > c + \varepsilon) \\ &\leq F_{Z_n}(c - \varepsilon) + 1 - F_{Z_n}(c + \varepsilon) \rightarrow 0. \end{aligned} \quad \blacksquare$$

Remark 3 (Chain of Implications). Let Z_n be a sequence of random variables and Z be a random variable or constant. Then we have the following chain of implications:

$$Z_n \xrightarrow{ms} Z \implies Z_n \xrightarrow{p} Z \implies Z_n \rightsquigarrow Z. \quad (6)$$

The converse does not hold in general. **Example 1** shows that the first implication is strict; And unless Z is a constant, convergence in distribution does not imply convergence in probability.

B Useful Theorems

Lemma 1 (Concentration around the mean). Let $\{Z_n\}$ be a sequence of random variables with finite variance for every n and $\mathbf{Var}(Z_n) \rightarrow 0$. Then

$$Z_n - \mathbb{E}(Z_n) \xrightarrow{p} 0. \quad (7)$$

If also $\mathbb{E}(Z_n) \rightarrow c$ for a constant c , then $Z_n \xrightarrow{p} c$.

Proof. Chebyshev's inequality gives, for every $\varepsilon > 0$,

$$\Pr(|Z_n - \mathbb{E}(Z_n)| > \varepsilon) \leq \frac{\mathbf{Var}(Z_n)}{\varepsilon^2} \rightarrow 0.$$

If $\mathbb{E}(Z_n) \rightarrow c$, then for sufficiently large n , $|\mathbb{E}(Z_n) - c| < \varepsilon/2$. Hence, we have

$$\begin{aligned} \Pr(|Z_n - c| > \varepsilon) &\leq \Pr(|Z_n - \mathbb{E}(Z_n)| + |\mathbb{E}(Z_n) - c| > \varepsilon) \\ &\leq \Pr(|Z_n - \mathbb{E}(Z_n)| > \varepsilon/2) \leq \frac{4 \mathbf{Var}(Z_n)}{\varepsilon^2} \rightarrow 0. \end{aligned} \quad \blacksquare$$

Theorem 1 (Weak Law of Large Numbers (WLLN)). Let $\{X_i\}_{i=1}^n$ be independent and identically distributed (iid) with mean μ and finite variance σ^2 . Then the sample mean $\bar{X}_n = n^{-1} \sum_{i=1}^n X_i$ is consistent for μ .

Proof. Apply Lemma 1 with $Z_n = \bar{X}_n$ and $c = \mu$. \blacksquare

Remark 4. The finite second moment assumption in Theorem 1 can be dropped. In fact, WLLN holds for iid random variables with finite first moment. That is, if $\{X_i\}_{i=1}^n$ is iid with $\mathbb{E}(X_i) = \mu$ finite, then $\bar{X}_n \xrightarrow{p} \mu$. However, the proof of this stronger result is more involved.

Theorem 2 (Continuous Mapping). Let $Z_n \in \mathbb{R}^k$ be random vectors and $Z \in \mathbb{R}^k$ be a random vector. Let g be a continuous function. Then we have

$$Z_n \xrightarrow{p} Z \implies g(Z_n) \xrightarrow{p} g(Z), \quad (8)$$

$$Z_n \rightsquigarrow Z \implies g(Z_n) \rightsquigarrow g(Z). \quad (9)$$

Proof. Omitted. \blacksquare

Example 3. If $(Z_{1n}, Z_{2n})' \xrightarrow{p} (Z_{1,0}, Z_{2,0})' \in \mathbb{R}^2$. By Theorem 2, we have

$$Z_{1n} + Z_{2n} \xrightarrow{p} Z_{1,0} + Z_{2,0}, \quad (10)$$

$$Z_{1n} Z_{2n} \xrightarrow{p} Z_{1,0} Z_{2,0}, \quad (11)$$

$$Z_{1n}/Z_{2n} \xrightarrow{p} Z_{1,0}/Z_{2,0} \quad (Z_{2,0} \neq 0) \quad (12)$$

by applying the continuous mapping theorem to the functions $g(x, y) = x + y$,

$g(x, y) = xy$, and $g(x, y) = x/y$ respectively to (Z_{1n}, Z_{2n}) .

Theorem 3 (Slutsky's Theorem). Let Z_n and W_n be sequences of random variables. If $Z_n \rightsquigarrow Z$ and $W_n \xrightarrow{p} c$ for a constant c , then

$$Z_n + W_n \rightsquigarrow Z + c, \quad Z_n W_n \rightsquigarrow cZ, \quad Z_n/W_n \rightsquigarrow Z/c \quad (c \neq 0). \quad (13)$$

Sketch of Proof. Convergence in probability implies $W_n \rightsquigarrow c$. One can show that $(Z_n, W_n)' \rightsquigarrow (Z, c)'$. Then apply **Theorem 2** to the functions $g(x, y) = x + y$, $g(x, y) = xy$, and $g(x, y) = x/y$ respectively to $(Z_n, W_n)'$. ■

Theorem 4 (Lindeberg-Lévy **Central Limit Theorem (CLT)**). Let $\{X_i\}_{i=1}^n$ be iid with mean μ and finite variance σ^2 . Then

$$\sqrt{n}(\bar{X}_n - \mu) \rightsquigarrow \mathcal{N}(0, \sigma^2). \quad (14)$$

Proof. Omitted. ■

Theorem 5 (Delta Method). Let $Z_n \in \mathbb{R}^k$ be a sequence of random vectors and $Z \in \mathbb{R}^k$ be a fixed vector. Let $g : \mathbb{R}^k \rightarrow \mathbb{R}^m$ be continuously differentiable in a neighborhood of Z , with Jacobian matrix $G(Z) \in \mathbb{R}^{m \times k}$. If $\sqrt{n}(Z_n - Z) \rightsquigarrow Z_0$, then

$$\sqrt{n}(g(Z_n) - g(Z)) \rightsquigarrow G(Z)Z_0. \quad (15)$$

Proof. Since $\sqrt{n}(Z_n - Z) = O_p(1)$, we have $Z_n - Z = O_p(n^{-1/2})$ and hence $Z_n \xrightarrow{p} Z$. Since $Z_n \xrightarrow{p} Z$, it suffices to work in the neighborhood where g is continuously differentiable. The integral form of the mean value theorem gives the matrix identity

$$g(Z_n) - g(Z) = H_n(Z_n - Z) \quad \text{where} \quad H_n := h(Z_n) := \int_0^1 G(Z + t(Z_n - Z)) dt.$$

Continuity of G makes h continuous at Z with $h(Z) = G(Z)$. Since $Z_n \xrightarrow{p} Z$, **Continuous Mapping** gives $h(Z_n) \xrightarrow{p} h(Z)$, i.e., $H_n - G(Z) = o_p(1)$. Hence, we have

$$\begin{aligned} \sqrt{n}(g(Z_n) - g(Z)) &= \sqrt{n}H_n(Z_n - Z) \\ &= G(Z)\sqrt{n}(Z_n - Z) + \underbrace{\sqrt{n}(H_n - G(Z))}_{o_p(1)} \underbrace{(Z_n - Z)}_{O_p(n^{-1/2})}. \end{aligned}$$

The first term converges in distribution to $G(Z)Z_0$ by the continuous mapping theorem, while the second converges in probability to zero. **Slutsky's Theorem** gives the stated result. ■